

AquaSecure

Technical Documentation

Complete platform reference — collection, pipeline, routing, core platform and AI agents

Version 1.0 · June 2026

Contents

This document brings together the five core component references into one complete technical reference. The parts follow the path data takes through the platform.

- Part 1** **Aqua SIEM** — Data and log collection layer
- Part 2** **AQUA Edge** — Observability pipeline and OCSF normalization
- Part 3** **AQUA Proxy** — Receiving and tenant routing layer
- Part 4** **Core Platform** — Storage, AI, enrichment and modules
- Part 5** **AI Agents** — SOC Agent Studio — the agent roster

Introduction

AquaSecure is an AI-native, modular cybersecurity platform. Security telemetry is collected at the source, normalized into a common schema, routed by tenant, and then stored, analyzed by AI and surfaced through governance and security modules. Each stage is a distinct component, documented as a part of this reference.

The platform is organized as an end-to-end pipeline:

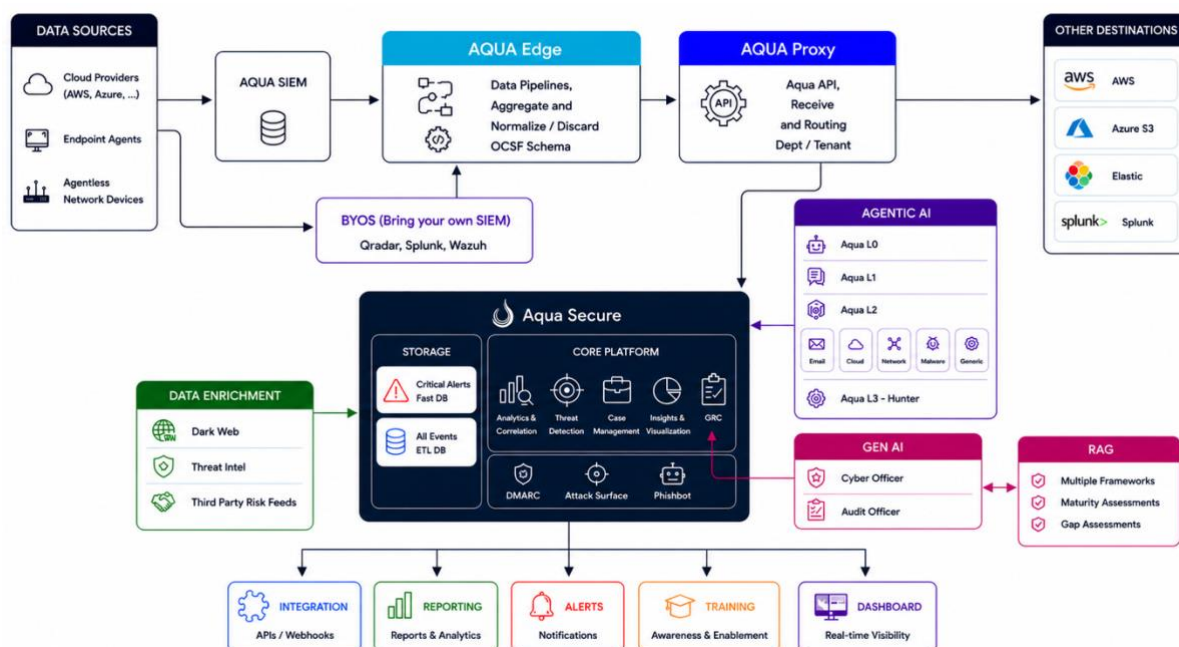


Figure 1 — End-to-end data flow across the AquaSecure platform.

How this document is organized

Each part is a self-contained component reference with its own sections and diagrams:

Part 1 — Aqua SIEM collects and aggregates data and logs from across the environment using stock Wazuh.

Part 2 — AQUA Edge is the observability pipeline that parses, filters, structures, secures and normalizes telemetry to OCSF; it can also ingest a customer's existing SIEM via BYOS.

Part 3 — AQUA Proxy receives data via webhooks and routes it by department and tenant to external destinations and the platform.

Part 4 — Core Platform stores the data across two pipelines, applies the three-tier AI and enrichment, and exposes the governance and security modules and the employee portal.

Part 5 — AI Agents documents the SOC Agent Studio roster: the L0–L3 tiers and supporting agents, with their triggers and priorities.

Aqua SIEM

Core Component — Technical Documentation

Data and log collection layer of the AquaSecure platform

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1. Overview

Aqua SIEM is the data and log collection layer of the AquaSecure platform. Its job is to gather security telemetry from across the environment, aggregate it, and hand it off to the rest of the platform. Aqua SIEM is built on Wazuh as its baseline collection technology.

2. Design Principle: Wazuh as Baseline

Aqua SIEM uses Wazuh as its underlying data-collection engine, in standard form. Wazuh agents are deployed as-is — they are not modified, patched or custom-built.

This is deliberate. By keeping the collection layer stock, the platform:

- **Tracks upstream releases without rework** — a new Wazuh version can be adopted without breaking anything in AquaSecure.
- **Lowers maintenance** — there is no forked or patched agent to maintain across upgrades.
- **Inherits upstream improvements** — fixes, performance gains and new platform support from Wazuh are available automatically.

3. Scope & Responsibilities

3.1 In scope

- **Collect:** deploy and operate stock Wazuh agents and collectors to gather data and logs.
- **Aggregate:** receive and consolidate telemetry from endpoints, network devices and cloud sources.
- **Hand off:** make the collected data available to the rest of the platform.

3.2 Out of scope

- **Downstream processing & analysis:** handled by other components, after Aqua SIEM.
- **Agent-side customization:** no custom logic is embedded in the agents, by design, to preserve upgrade safety.

4. Data Collection

Aqua SIEM collects from three source types, all through standard Wazuh mechanisms:

- **Endpoints** — servers and workstations running stock Wazuh agents that forward logs and events. Collection only, not endpoint detection.
- **Network devices** — firewalls, routers and switches collected agentlessly (e.g. via syslog).
- **Cloud sources** — logs from AWS, Azure and similar platforms via cloud log integration.

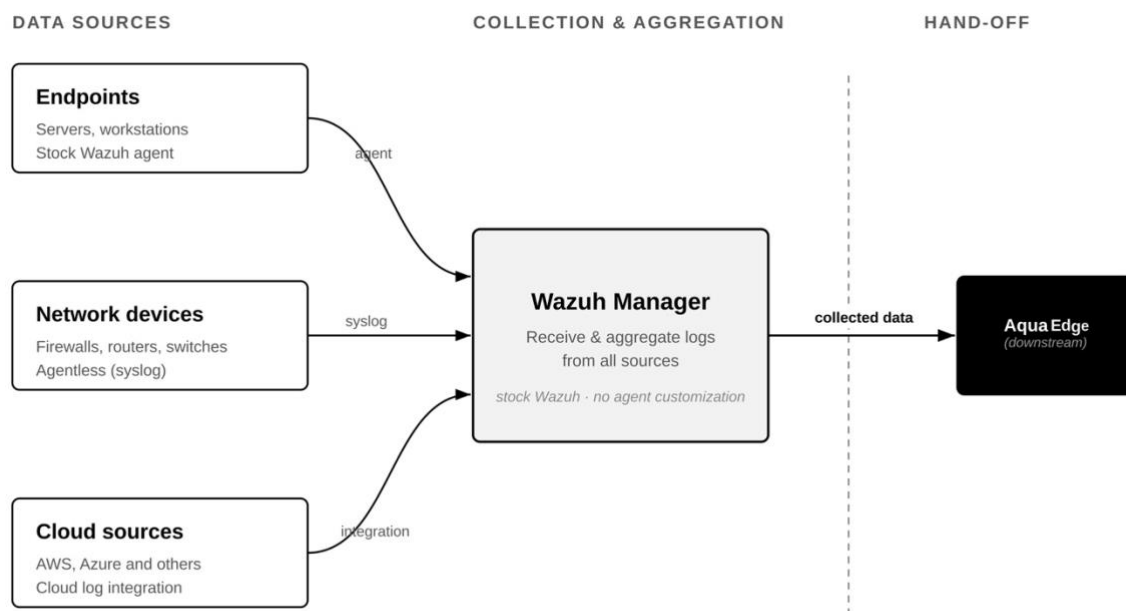
Collection behaviour is driven through standard Wazuh configuration rather than agent code changes, so the collection layer remains upgrade-safe.

5. Version & Upgrade Resilience

Because no Wazuh agent is customized or forked, future Wazuh versions are expected to work without changes to AquaSecure. The collection layer sits behind a clean hand-off, so upgrading Wazuh does not ripple into the rest of the platform. The practical result is lower upgrade risk, reduced maintenance overhead, and faster adoption of upstream Wazuh releases.

6. Collection Flow

Data and logs are collected from endpoints, network devices and cloud sources, aggregated at the Wazuh manager, and handed off to the platform. Everything up to the hand-off is stock Wazuh.



Aqua SIEM scope: collect and aggregate. Everything left of the hand-off is stock Wazuh, kept upgrade-safe.

Figure 1 — Aqua SIEM data/log collection flow.

AQUA Edge

Core Component — Technical Documentation

Observability pipeline and normalization layer of the AquaSecure platform

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1. Overview

AQUA Edge is the observability pipeline and normalization layer of the AquaSecure platform — designed both for pipeline solutions and for normalization. At its core is a high-performance, expression-based scripting language built into the pipeline that acts as its “brain.” Sitting between data intake and data routing, it parses, filters, structures and secures telemetry in real time, and normalizes it to a common schema (OCSF). It can also ingest from a customer’s existing SIEM through Bring Your Own SIEM (BYOS).

2. Role in the Platform

AQUA Edge sits between intake and routing. It receives telemetry from a wide range of input sources — including data handed off by the collection layer — transforms and normalizes it, and routes the result to a wide range of output sinks (storage, search and analytics engines, monitoring platforms, the core platform, and more). It is the stage where raw, heterogeneous telemetry becomes clean, consistent, OCSF-normalized data.

3. The Pipeline Engine (the “Brain”)

The engine is expression-based and high-performance, operating on data in real time as it flows through. Its responsibilities are:

- **Parse** — interpret raw and unstructured telemetry into structured fields.
- **Filter** — drop noise, sample, and select events to control volume and cost.
- **Structure** — reshape, enrich and standardize event structure.
- **Secure** — redact or mask sensitive fields and protect telemetry as it moves.
- **Normalize** — map events to the OCSF schema for a consistent downstream data model.
- **Route** — direct each processed event to the appropriate output sink.

4. Input Sources

AQUA Edge accepts telemetry from seven categories of source:

Input Source	What it ingests
Files & System Streams	Local text logs, system daemon events, and system process outputs.
Containers & Orchestration	Logs from container runtimes and cluster orchestration engines.
Network Sockets	Raw streaming data over standard transport protocols (TCP, UDP, HTTP).
Message Brokers	Streaming messages from distributed queues, publish-subscribe clusters, and key-value datastores.
Cloud Infrastructure Services	Log events, metrics, and storage object streams from public cloud providers.

Input Source	What it ingests
Pipeline Agents	Forwarded data streams from upstream collectors and secondary pipeline instances.
Customer SIEM (BYOS)	Data from a customer's existing SIEM, accepted through the Bring Your Own SIEM (BYOS) model.

5. Bring Your Own SIEM (BYOS)

AQUA Edge can also receive data from a customer's existing SIEM through the Bring Your Own SIEM (BYOS) model.

Business use case: where a customer already operates a SIEM for data collection, there is no need to replace it. AQUA Edge accepts data directly from the existing SIEM and takes it into the pipeline for processing, normalization and routing alongside every other source — so the customer keeps their current collection investment while gaining AQUA Edge's processing and OCSF normalization.

BYOS receives data from following SIEM: Splunk, QRadar, Wazuh and Elastic

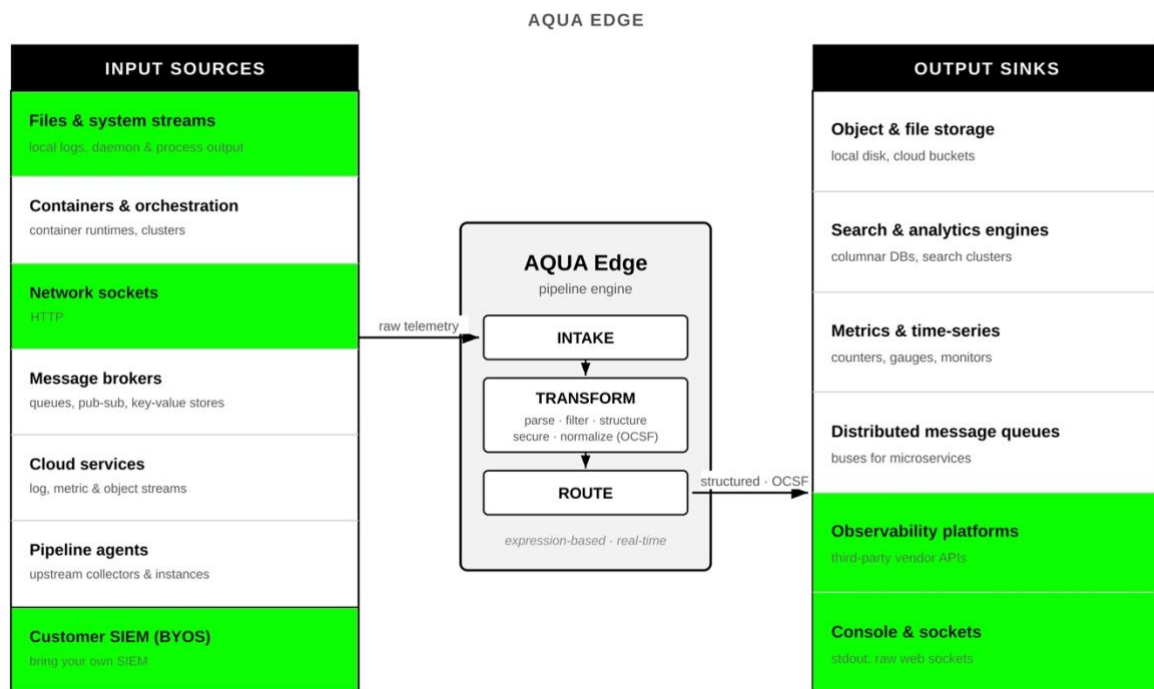
6. Output Sinks

After processing, data can be routed to six categories of destination:

Output Sink	What it writes / forwards
Object & File Storage	Structured telemetry written to local disk files or long-term cloud object storage buckets.
Search & Analytics Engines	Processed logs forwarded to columnar databases and indexing search clusters for querying.
Metrics & Time-Series Platforms	Calculated numbers, counters, and gauges routed to dedicated monitoring databases.
Distributed Message Queues	Cleaned telemetry published back into message buses for downstream microservices.
Managed Observability Platforms	Unified logs, metrics, and traces sent to third-party vendor APIs.
Network & Terminal Consoles	Data printed to standard output or streamed over raw web sockets.

7. Pipeline Flow

Telemetry enters from any combination of input sources, passes through the engine — intake, then transform (parse, filter, structure, secure, normalize to OCSF), then route — and is delivered to one or more output sinks.



AQUA Edge sits between intake and routing — parsing, filtering, structuring, securing and normalizing telemetry to OCSF in real time.

Figure 1 — AQUA Edge pipeline flow: intake → transform (parse, filter, structure, secure, normalize) → route.

AQUA Proxy

Core Component — Technical Documentation

Receiving and routing layer of the AquaSecure platform

Version 1.0 · June 2026

1. Overview

AQUA Proxy is the receiving and routing layer of the AquaSecure platform. It receives data delivered through webhooks, routes it by department and tenant, and forwards it to its destinations — both external systems and the AquaSecure platform itself.

2. Role in the Platform

AQUA Proxy is the platform's tenant-aware distribution point. Incoming data arrives via webhooks; AQUA Proxy determines the correct department or tenant for each payload and forwards it onward. From here, data is directed both to external destinations and into the AquaSecure platform, where it feeds storage, analytics and the rest of the core.

3. Responsibilities

- **Receive** — accept data delivered via webhooks.
- **Route** — direct each payload to the correct department or tenant.
- **Distribute** — forward routed data to external destinations and to the AquaSecure platform.

Filtering is not part of AQUA Proxy; data is routed as received.

4. Receiving (Webhooks)

AQUA Proxy receives data through webhooks — HTTP callbacks that deliver payloads as events occur upstream. Incoming payloads are accepted at the proxy and passed directly to routing, with no filtering or transformation applied at this layer.

5. Department / Tenant Routing

Each payload is routed to the appropriate department or tenant. This tenant-aware routing is what allows a single deployment to serve multiple departments — or, in a partner model, multiple customer organizations — while keeping each tenant's data directed correctly and kept apart.

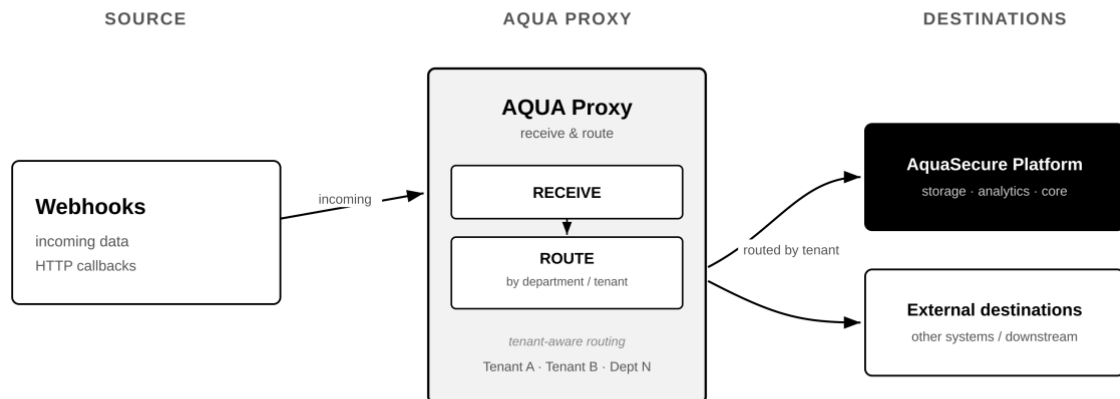
6. Destinations

Routed data is distributed to two kinds of destination:

- **External destinations** — other systems and downstream targets outside the platform.
- **AquaSecure platform** — where the data feeds storage, analytics and the core platform.

7. Routing Flow

Data arrives from webhooks, is received and routed by department and tenant, and is then distributed to external destinations and the AquaSecure platform.



AQUA Proxy receives data from webhooks, routes it by department and tenant, and distributes it to external destinations and the AquaSecure platform.

Figure 1 — AQUA Proxy routing flow: receive via webhooks → route by tenant → distribute.

Core Platform

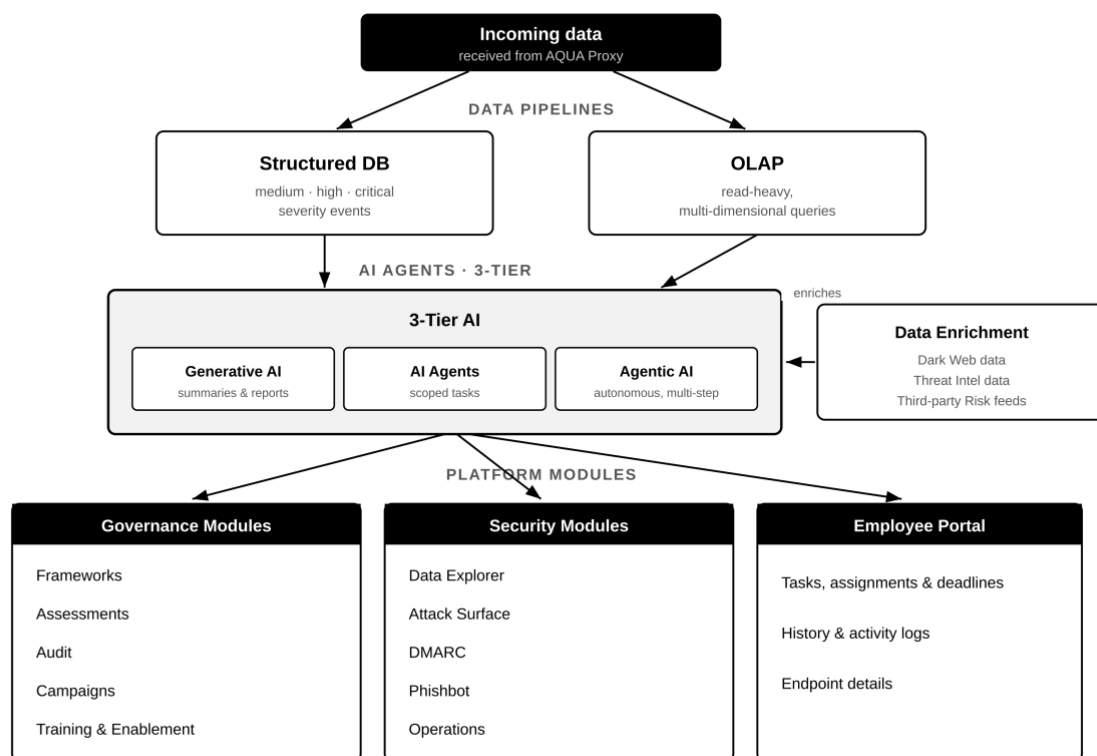
Core Component — Technical Documentation

Storage, analysis, enrichment and modules of the AquaSecure platform

Version 1.0 · June 2026

1. Overview

The Core Platform is where data received into AquaSecure is stored, analyzed, enriched and acted upon. Incoming data first lands in two data pipelines; it is then analyzed by a three-tier AI layer and enriched with external intelligence; and it is finally surfaced through the platform's governance and security modules and an employee portal.



Data lands in two pipelines (Structured DB / OLAP), is analyzed by the 3-tier AI and enriched, then surfaced through the platform modules.

Figure 1 — Core Platform architecture and data flow.

2. Data Pipelines

When data is received, it first hits two data pipelines:

- **Structured DB** — AQUA Edge identifies the severity of each event; everything above low severity (medium, high and critical) is stored here for fast, structured access.
- **OLAP (Online Analytical Processing)** — the remaining data is held here for read-heavy, multi-dimensional analytical queries.

3. AI Agents — 3-Tier AI

From the pipelines, data is analyzed by the AI layer. AquaSecure combines three types of AI, used together:

- **Generative AI** — produces human-readable output such as summaries, explanations and reports from security data.
- **AI Agents** — perform specific, scoped tasks on request.
- **Agentic AI** — operate autonomously across multiple steps, chaining actions to investigate and respond.

4. Data Enrichment

The platform enriches its data and analysis with external intelligence feeds:

- **Dark Web data**
- **Threat Intelligence data**
- **Third-party Risk feeds**

5. Governance Modules

The governance modules deliver governance, risk and compliance (GRC) capability.

5.1 Frameworks

Maps business operations directly to international security standards and regulatory guidelines such as ISO 27001, SOC 2 and GDPR. It helps establish corporate policies, manage internal audits, and maintain compliance through automated control tracking, and provides templates for the policies, documents and SOPs each framework requires. All frameworks are cross-mapped, so controls already satisfied for one framework are automatically identified against the others — reducing duplicate work.

5.2 Assessments

A set of assessments that establish where the organization stands:

- **Cyber Maturity** — establishes which controls are already implemented (an important first step before GRC).
- **Gap Assessment** — checks whether the organization has all the required documents and processes.
- **Readiness** — a pre-compliance stage where everything is attached and mapped, so that when the compliance audit begins it maps automatically.

5.3 Audit

Provides the ability to audit the organization, and to continuously monitor in preparation for the next audit.

5.4 Campaigns

Lets internal employees review and agree to newly implemented policies and SOPs. For example, when a new policy is assigned to employees, they can formally acknowledge it.

5.5 Training & Enablement

Compliance training and tracking of employee policy agreement in one central location. It delivers bite-sized security lessons, manages mandatory certifications, and provides completion records to evidence regulatory compliance during security reviews.

6. Security Modules

The security modules provide traditional monitoring of cyber threats.

6.1 Data Explorer

A central place to search, visualize and investigate security alerts and raw logs from all data sources.

6.2 Attack Surface

Discovers, monitors and secures internet-connected business assets before attackers can exploit them. It continuously scans the digital footprint to find hidden vulnerabilities, exposed servers, open ports and forgotten websites — stopping breaches before they start.

6.3 DMARC

Protects email domains from being spoofed or hijacked. It sets up domain rules that automatically block fake emails, help legitimate messages reach client inboxes, and provides clear reporting dashboards to monitor email security.

6.4 Phishbot

Sends safe, automated simulated phishing emails to test and train employees against attacks. It tracks who is caught, immediately teaches them how to spot the threat, and provides dashboards to measure security improvement over time.

6.5 Operations

A bird's-eye view where information from across the platform comes together. It includes incident management, alert ticketing system, case management and asset management. Initial detection and investigation are performed by the AI; Once the AI has detected and investigated an event, resolution is carried out by the security team, with Operations keeping each event tracked through to closure.

7. Employee Portal

The employee portal is for the platform's users and employees. From here, an employee can view the tasks, assignments and deadlines assigned to them from across the entire platform, along with their own history and activity logs, and the endpoint details of their machines.

AI Agents

SOC Agent Studio — Technical Documentation

The agent roster: tiers, triggers, priorities and behaviour

Version 1.0 · June 2026

1. Overview

SOC Agent Studio is where AquaSecure's SOC agents are built, configured and deployed — composed by dragging blocks onto a canvas and published to activate. The agents form a tiered detection-and-investigation pipeline: an L0 pre-processing hook, an L1 router, a set of L2 domain specialists, an L3 threat hunter, and a group of supporting agents for IOC validation, reporting and scheduled hunting.

There are currently 14 agents. This document describes the agent model, how agents are triggered and ordered, and the role of each agent.

2. Agent Model

- **Lifecycle** — every agent moves through draft → publish → activate. Agents are composed from blocks on the Studio canvas.
- **Triggers** — pre (pre-create hook, runs before a record is created), post (post-create hook, runs after), and scheduled (cron, runs on a fixed cadence such as */15 = every 15 minutes).
- **Priority (p1–p100)** — sets execution order in the per-alert pipeline; lower numbers run earlier (p1 → p10 → p20 → p30 → p100).
- **Tenant scope** — some agents run across all tenants (shown as “6 tenants”), including L0, L1 and L3.
- **Versioning & status** — each agent is versioned (e.g. v1, v8) and is either published or draft.

3. Execution Pipeline

Within a single alert, agents run in priority order: L0 pre-processing, L1 routing, the matched L2 specialist (or the generic fallback), and finally the supporting agents. Separately, scheduled agents run on a cron cadence over raw SIEM data, independent of any individual alert.

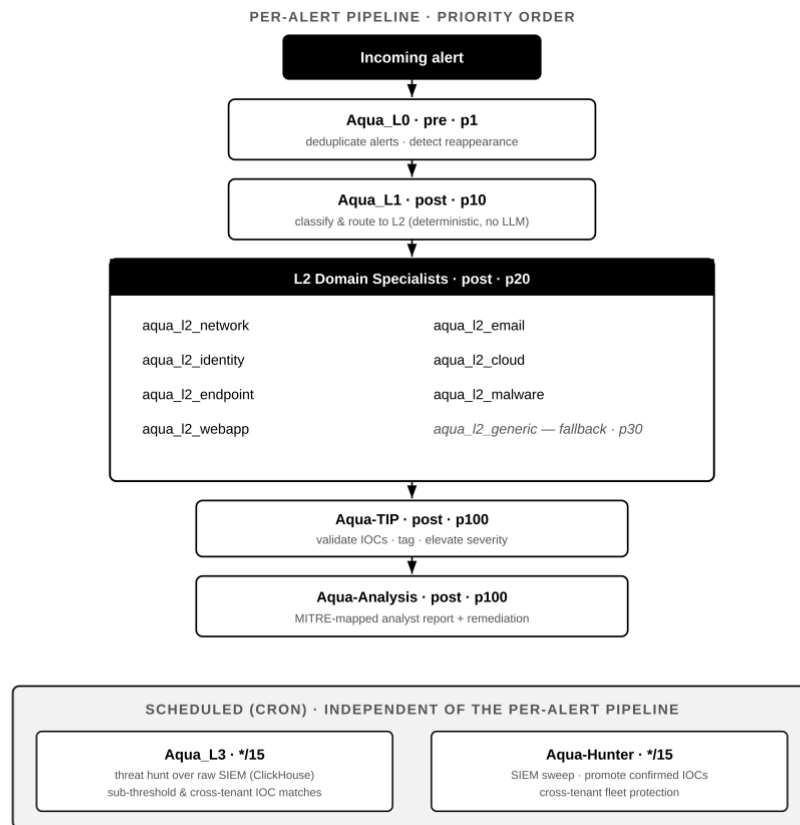


Figure 1 — Agent execution: per-alert priority pipeline and scheduled agents.

4. Agent Catalog

All 14 agents, in execution order:

Agent	Trigger · Prio	Purpose
Aqua_L0	pre · p1	Dedup alerts by fingerprint; detect reappearance of human-closed threats.
aqua_l1	post · p10	Classify and route each alert to the correct L2 specialist (deterministic).
aqua_l2_network	post · p20	Network-layer investigation; IP/domain TI checks; ClickHouse pivot on src_ip ±24h.
aqua_l2_identity	post · p20	Auth, logon, RBAC and credential attacks; ClickHouse pivot on username ±2h.
aqua_l2_endpoint	post · p20	Host events (FIM, rootkit, processes, anomalies); ClickHouse pivot on hostname.
aqua_l2_webapp	post · p20	Web-app attacks (WAF / app FW / web logs); attacker-IP TI + ClickHouse pivot.
aqua_l2_email	post · p20	Phishing URL analysis and email header investigation.
aqua_l2_cloud	post · p20	Cloud alerts: GuardDuty / Azure Sentinel / GCP SCC.
aqua_l2_malware	post · p20	Hash IOC validation via VirusTotal sandbox + YARA rules.
aqua_l2_generic	post · p30	Fallback analyst when no specialist fits; uses the primary LLM.
aqua_l3	cron */15	Scheduled threat hunt over raw SIEM for sub-threshold / cross-tenant threats.
Aqua-TIP	post · p100	Validate IOCs against AquaTIP; tag and elevate severity.

Agent	Trigger · Prio	Purpose
Aqua-Analysis	post · p100	Scheduled MITRE-mapped analyst report from TIP/Hunter enrichment.
Aqua-Hunter	post · p100	Scheduled SIEM sweep; promote confirmed IOCs; cross-tenant protection.

5. Agents by Tier

5.1 L0 — Pre-processing

Aqua_L0 (pre · p1 · all tenants · published) — A pre-create hook that runs on every incoming alert before anything else. It deduplicates repeated alerts by fingerprint — grouping them under one parent — and detects when a human-closed threat reappears.

5.2 L1 — Triage & Routing

aqua_l1 (post · p10 · all tenants · published) — Classifies every incoming alert and routes it to the correct L2 domain specialist (Network, Identity, Endpoint, WebApp). Routing is deterministic — based on keyword sets and IOC type, with no LLM call.

5.3 L2 — Domain Specialists

Each specialist performs a deep investigation for its domain (post · p20, published):

aqua_l2_network (post · p20) — Network-layer alerts. Validates attacker IPs and domains against 7 threat-intelligence providers, then pivots ClickHouse SIEM events on the attacker source IP within a ± 24 h window.

aqua_l2_identity (post · p20) — Authentication failures, Windows logon events, RBAC violations and credential-based attacks. Pivots ClickHouse on the affected username(s) within a ± 2 h window to classify the attack.

aqua_l2_endpoint (post · p20) — Host-based security events: file integrity monitoring, rootkit detection, suspicious processes, network anomalies and software changes. Pivots ClickHouse on the affected hostname.

aqua_l2_webapp (post · p20) — Web application attacks detected by WAF, application firewalls or web server logs. Validates the attacker IP against threat-intelligence providers, then pivots ClickHouse.

aqua_l2_email (post · p20) — Phishing URL analysis and email header investigation.

aqua_l2_cloud (post · p20) — Cloud alert investigation across GuardDuty, Azure Sentinel and GCP SCC.

aqua_l2_malware (post · p20) — Hash IOC validation via VirusTotal sandbox and YARA rules.

aqua_l2_generic (post · p30 · fallback) — Fallback SOC analyst. Runs when no specialist (network / identity / endpoint / webapp) clearly fits the alert. Uses the primary LLM (symlink:8b by default, via the primary-llm alias) for general analysis.

5.4 L3 — Threat Hunting

aqua_l3 (cron */15 · all tenants · published · v8) — A scheduled threat hunt that sweeps raw SIEM events (ClickHouse) on a cadence for threats that never crossed the alert-creation threshold — low-severity events and cross-tenant IOC matches.

5.5 Supporting Agents

Aqua-TIP (post · p100 · published) — Validates IOCs (IPs, domains, hashes, URLs, CVEs) against AquaTIP via a cache + provider chain. Tags the alert with tipTags and iocMatches, and elevates severity if malicious.

Aqua-Analysis (post · p100 · draft) — A scheduled narrative writer (roughly every 20 minutes). Reads TIP/Hunter enrichment and produces a structured, MITRE-mapped analyst report with remediation steps.

Aqua-Hunter (post · p100 · draft) — A scheduled SIEM sweep (roughly every 15 minutes). Promotes vendor-confirmed malicious indicators into the cache and provides cross-tenant fleet protection.